

Calvin Yang

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PROFESSIONAL SUMMARY

Computer Science and Engineering student at UC Irvine with hands-on experience in embedded systems, firmware development, and hardware design. Background spans C/C++ firmware, microcontroller integration, electromagnetic simulation, and low-level circuit work across independent and team engineering projects.

EDUCATION

University of California, Irvine
Bachelor of Science, Computer Science and Engineering
Northwood High School
GPA: 3.77/4.00

Irvine, CA
Expected Jun. 2030
Irvine, CA
Aug. 2023 – Jun. 2026

PROJECTS

Electromagnetic Accelerator (Coil Gun)

Jun. 2022 – Present

Individual Project

- Designed and built a two-stage coil gun, solving the challenge of precisely timing sequential electromagnetic coils to efficiently accelerate a projectile.
- Used ANSYS Maxwell 2D for electromagnetic field simulation and coil-timing optimization, Fusion 360 for mechanical design, 3D printing for fabrication, and Arduino for stage-control logic.
- Achieved approximately 16 m/s projectile velocity with a 54.63 g projectile using four 400 V, 330 μ F capacitors.

Arduino Sensor & Control System

Jun. 2026 – Jul. 2026

Individual Project

- Built a real-time sensor monitoring system using an Arduino Uno R3, integrating an ultrasonic distance sensor and temperature/humidity sensor with an OLED display.
- Developed C/C++ firmware to collect sensor measurements and refresh the OLED display at 50 ms intervals.
- Implemented I2C communication between the Arduino, sensors, and OLED display.

EXPERIENCE

UCI Master of Engineering Capstone Project Assistant

Jan. 2025 – Jun. 2025

Gaudi Technology

Irvine, CA

- Revised 3D CAD models in Fusion 360 for printable concrete structures.
- Assisted with fabrication and preparation of concrete molds used for ASTM material strength testing.
- Collaborated with 4 MoE student researchers during prototype development and testing.

ACTIVITIES & LEADERSHIP

American Rocketry Challenge

Nov. 2025 - Mar. 2026

R&D and Simulation Lead

- Improved simulated flight accuracy by 23.2% through iterative OpenRocket analysis.
- Collaborated with a 7-member engineering team to design and build a 25.6-inch competition rocket.
- Completed two competition qualification flights, achieving an official score of 139.8 total.

YouTube (youtube.com/@SomeHeresyGaming)

Jan. 2026 - Present

Content Creator

- Built and grew a YouTube channel to 50,000+ subscribers and 150M+ total views.

SKILLS

Technical: C/C++, Embedded Systems, Circuit Analysis, Java, Python, Soldering, 3D Modeling, Video Editing

Tools: ANSYS Maxwell, EasyEDA, Arduino (Uno R3), Fusion 360, OpenRocket, VS Code, CapCut